

MARION REEF, Australia

WMO: 942980

Lat: 19.096S

Lon: 152.389E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.1	20.8	N/A	N/A	N/A	N/A	N/A	N/A	14.2	22.1	13.5	22.3	8.2	170	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.9	29.6	N/A	29.2	N/A	28.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
14.3	13.3	12.4	DB	18.7	31.6	1.0	1.8	18.0	32.9	17.4	33.9	16.8	34.9	16.1	36.2	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	27.3	27.6	27.1	26.0	24.7	23.5	22.6	22.5	23.2	24.3	25.4	26.7
	DBStd	2.02	0.95	0.87	0.92	0.87	0.96	1.04	0.94	0.82	0.91	0.98	0.83	0.93
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5492	536	492	529	479	457	404	391	387	395	442	463	517
	CDD18.3	2450	278	259	271	229	198	154	132	128	145	183	213	259
	CDH23.3	17699	3040	2949	2884	2012	1158	413	109	60	214	759	1548	2553
	CDH26.7	2428	649	757	543	98	17	1	0	0	1	4	41	319
(14) Wind	WSAvg	7.7	7.1	7.1	8.3	8.9	8.4	8.4	8.2	7.6	7.1	7.2	7.2	6.9
(15) Precipitation	PrecAvg	877	140	171	165	78	55	52	39	30	14	21	34	78
	PrecMax	1402	419	399	515	325	228	200	217	119	49	144	227	314
	PrecMin	424	6	32	15	12	6	6	2	3	0	0	3	5
	PrecStd	257	104	103	140	72	51	42	49	30	12	30	49	67
	PrecMax	1402	419	399	515	325	228	200	217	119	49	144	227	314
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	30.6	29.7	28.7	28.1	26.5	25.3	25.0	26.0	27.0	28.8	29.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	29.5	29.7	29.1	28.0	27.1	25.7	24.7	24.4	25.3	26.2	27.7	28.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	29.0	29.2	28.8	27.6	26.5	25.2	24.3	24.0	24.8	25.8	27.1	28.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	28.6	28.8	28.5	27.2	26.1	24.9	24.0	23.6	24.4	25.5	26.6	28.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	1.7	1.9	1.7	1.5	1.4	1.5	1.4	1.4	1.4	1.4	1.5	1.5
		MCDBR	2.1	2.3	2.1	1.9	2.1	1.7	1.7	1.5	1.7	1.8	2.4	2.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.386	0.391	0.391	0.368	0.354	0.360	0.336	0.340	0.357	0.371	0.376	0.378
	taud		2.601	2.598	2.586	2.629	2.625	2.599	2.647	2.624	2.549	2.535	2.556	2.595
	Ebn at Noon		959	943	920	904	881	855	891	919	938	950	961	967
	Edh at Noon		105	104	101	91	85	84	82	89	103	109	109	105
(44) All-Sky Solar Radiation	RadAvg		6.94	6.50	5.77	5.27	4.49	3.95	4.31	5.27	6.36	7.07	7.39	7.44
	RadStd		0.50	0.61	0.55	0.32	0.28	0.29	0.23	0.33	0.20	0.35	0.60	0.52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page